Sp 25

1)a) ~~ø~~ In the 8086 microprocessor, the physical address (PA) is obtained by:

PA = (Segment register × 10) + offset register

* Segment register hold the starting address of a memory segment.

* Offset register holds the distance (offset) from segment base address

* ~~This~~ The two are combined to generate a 20-bit physical address.

PA of 1000 : 2000

PA = 1000H×10 + 2000H
   = 10000H + 2000H
   = 12000H

1b) & i) CISC (Complex Instruction set Computer)

Architecture where a single instruction can perform multiple low-level operations such as loading from memory, arithmetic, and storing to memory.

ii) RISC (Reduced Instruction Set Computer)

Architecture with simple instructions designed for faster execution, usually completing in a single clock cycle.

iii) ALU (Arithmetic Logic Unit)

Part of the CPU that performs arithmetic operations (additions, subtraction, multiplication, division) and logical operations (comparison, bit shifting, logical tests).

c) Steps in the Execution Cycle

1) Fetch:

* The program counter (PC) places the address of the next instruction into the Memory Address Register (MAR).

* The MAR sends this address onto the address bus and triggers a read signal.

* The instruction from memory is placed on the data bus, loaded into the Memory Buffer Register (MBR),

and then copied to the Instruction Register (IR).

2) Decode:

* The Control Unit (CU) examines the instruction in the IR.

* The instruction is decoded to determine the required operation.

* The CU prepares the processor components (ALU, registers, buses) to execute the instruction.

3) Execute:

* The Arithmetic Logic Unit (ALU) or other execution units carry out the instruction.

* The required operation (arithmetic, logic, data transfer) is performed, and results ~~and~~ are stored in registers or memory.

Q2a)

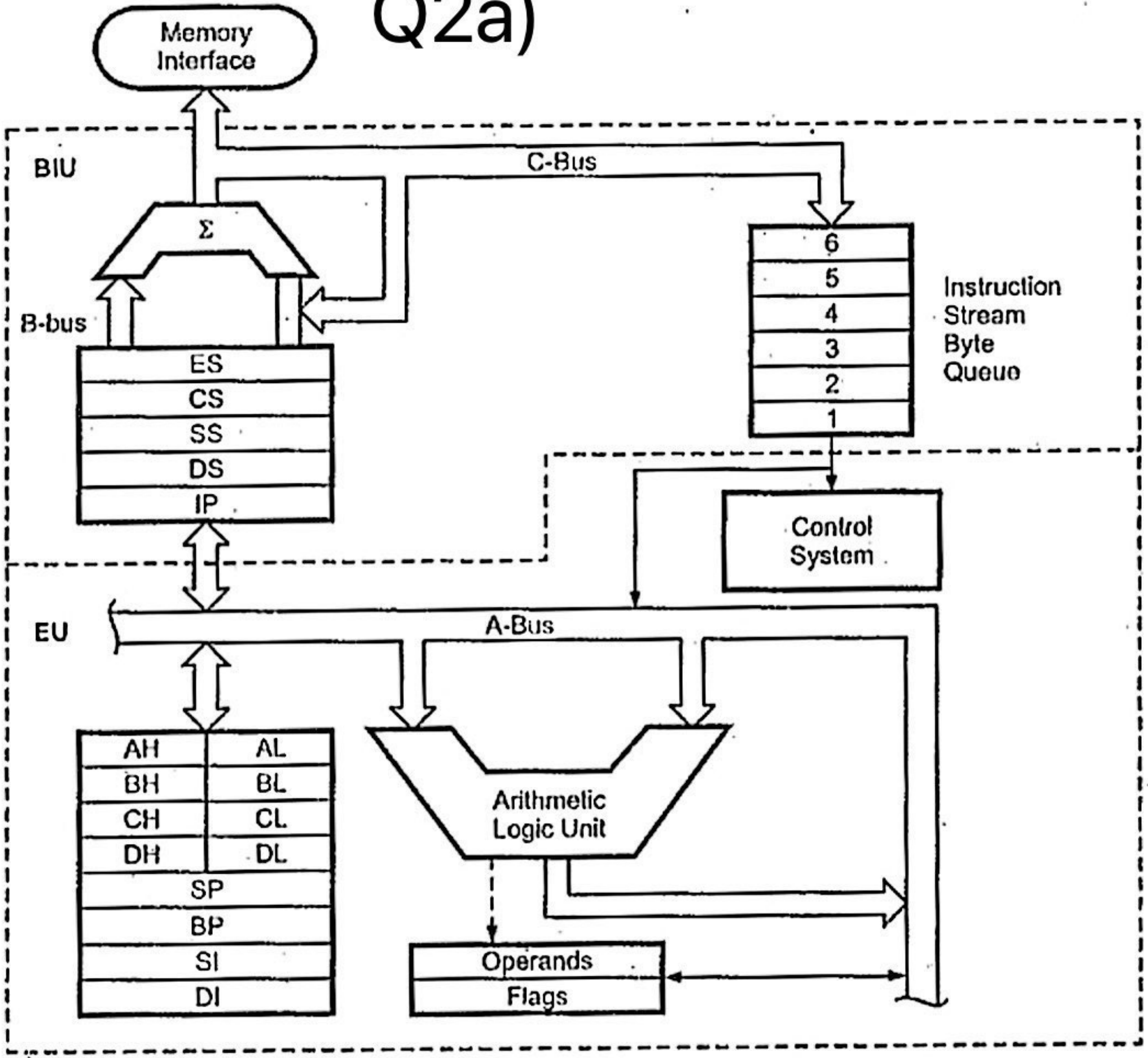

Fig. 6.2 8086 Internal block diagram

Sp 25

MOV AL, BL

2b) i) Addressing mode : Register addressing mode

Correct instruction → both AL and BL are 8 bit registers.

ii) MOV AL, BX

Addressing Mode : Register addressing mode.

Incorrect instruction → AL is 8-bit and BX is 16-bit

Correct form : MOV AL, BL (8-bit to 8-bit) or MOV AX, BX (16-bit to 16-bit).

iii) MOV AH, [BX]

Addressing mode : Register indirect addressing mode.

Correct instruction

iv) MOV CL, 3A8H

Addressing mode : Immediate addressing mode

Incorrect → CL is 8-bit, but 3A8H is ~~10 bits~~ 12-bit

Correct form : MOV CX, 3A8H (since CX is 16-bit)

2^n

n = no of bits

3A8H

$3\times16^2+10\times16^1+8\times16^0$

$= 936$ (decimal)

$2^9 = 512 < 936$

$2^{10} = 1024 > 936$

So 3A8H requires 10 bits

2c) Assembly Program:

```
MOV  AX, 20F1H
MOV  BX, 3C72H
ADD  AL, BL
ADC  AH, BH
```

* The ADD instruction adds the lower 8-bit parts (AL+BL)
* If there is a carry generated from this addition, the ADC instruction adds the upper 8-bit parts ~~(AH+BH)~~ (AH+BH+carry)
* This ensures correct 16-bit addition using two 8-bit operations.

Sp 25

2c) or)

```
MOV AL, 3FH ; AL = 3FH
MOV BL, 2AH ; BL = 2AH
ADD AL, BL    AL = 3FH + 2AH
```

$$
\begin{array}{rrl}
 & & 0111\ 11 \\
 & 3FH = & 0011\ 1111 \\
(+) & 2AH = & 0010\ 1010 \\
\hline
 & & 0110\ 1001
\end{array}
$$

$C=0$, $A=1$, $P=1$, $Z=0$, $S=0$, $O=0$

Sp 25

3)a) <u>Advantages of low level language</u>

* Programs developed using LLL are fast and memory efficient.
* No need of compiler or interpreter to translate the source to machine code.

<u>Disadvantages of low level language</u>

* Programs developed using LLL are machine dependent and not portable.
* Difficult to develop, ~~be to~~ debug and maintain

b) Assembly Language Program

	FF0EH	1111	1111	0000	1110
OR	0007H	0000	0000	0000	0111
		1111	1111	0000	1111
AND	3FFFH	0011	1111	1111	1111
		0011	1111	0000	1111
XOR	01A0H	0000	0001	1010	0000
		0011	1110	1010	1111

F F 0 E

15 15 0 14

1111 1111 0000 1110

Assembly Language Program :

```
MOV  AX, FF0EH
 OR  AX, 0007H
AND  AX, 3FFFH
XOR  AX, 01A0H
```

3(c) Assembly Program:

Instruction	AX	BX	SP
MOV AX, 8877H	8877	0000	0000
MOV BX, 79ABH	8877	79AB	0000
MOV SP, 2009H	8877	79AB	2009
PUSH AX	8877	79AB	2007
PUSH BX	8877	79AB	2005
POP AX	79AB	79AB	2007
POP BX	79AB	8877	2009